

Data Quality - Notes

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Tasks for Exercise 01: Introduction

Task 01

Data quality describes data characteristics at the meta-level. High-quality data provides a reliable and useful structure for information. Conversely, poor data quality can lead to mistrust and inaccurate outcomes.

Note: Personal definition.

Task 02

Scenario 2: E-Commerce – Inventory Management

How would incomplete data affect decision-making in your scenario?

Incomplete data can lead to a misunderstanding of consumer behavior and purchasing patterns. Without a full picture, businesses may fail to identify emerging trends or customer needs.

What could go wrong if the data in your scenario is not accurate?

Inaccurate data can result in supply chain bottlenecks or an oversupply of products. This leads to either lost sales due to stockouts or increased storage costs due to excess inventory.

How could the fact, that data is not available when required, affect your scenario?

A lack of real-time data availability undermines trust in data-driven processes. If stakeholders cannot access information when needed, they may revert to “gut-feeling” decisions, which are prone to error.

In what ways does the data need to be reliable and relevant?

To support effective inventory management, data must meet specific quality standards, such as:

- Availability: Data must be accessible to decision-makers at all times.
- Velocity: Data must be processed and updated at the speed of the business.
- Completeness: No critical data points (like regional demand) should be missing.
- Usability: Data must be in a format that is easy to interpret and act upon.

Note: The answers are not disjunct.

Tasks for Exercise 02: Data Quality Dimensions

Task 01

id	last_name	first_name	age	department	function	salary	commision_rate
1	Smith	Bill	56	Sales	Head of Sales	120.000	15%
2	Muller	John	25	Social Media	Creative Director	100.000	N/A
3	Grey	Anna	37	SEO	Google Expert	90.000	N/A
4	Berger	Lia	22	NULL	Freelancer		NULL
5	?	Mike	46	Facility	Team Manager	75.000	N/A
6	Doe	Jane	30	IT	Dev	N/A	NaN

- Scenario 1: The value exists but is not known (that is, known unknown): **last_name** of Mike (ID 5). Every person has a last_name, but it is currently missing from the dataset.
- Scenario 2: The value does not exist at all: **department** of Lia (ID 4). As an external freelancer, she is not part of the internal organizational structure.
- Scenario 3: The existence of the value is not known (that is, unknown unknown): **salary** of Jane (ID 6). It is unclear if she is a paid employee or an unpaid volunteer/intern; thus, the existence of the attribute itself is in question.

- Scenario 4: The attribute is not applicable: `commision_rate` for non-sales employees. This metric is only defined for sales roles and is fundamentally inapplicable to other functions.
- Scenario 5: The value is only populated under specific conditions: `salary` of Lia (ID 4). As a freelancer, the field remains empty until a specific hourly-based invoice or contract condition triggers the entry.

Tasks for Exercise 03: Exploratory Data Analysis

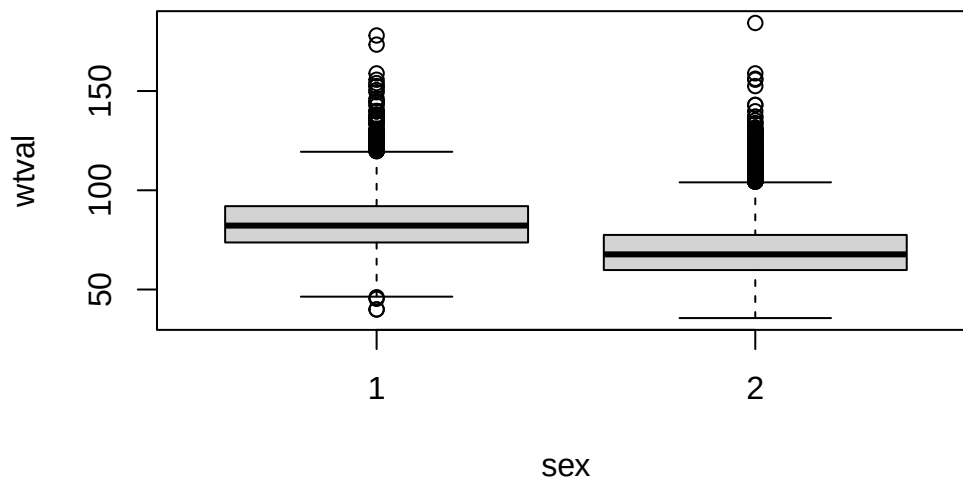
Task 01

Using R, generate a boxplot broken down by gender (variable `sex`). How can the boxplot be interpreted?

```
library(readxl)

data <- read_excel("/home/nils/dev/mscids-notes/fs26/dq/data/HSE.xlsx")

boxplot(wtval ~ sex, data = data)
```



We can clearly see a difference between the two classes.

Task 02

What are the distributions of the following boxplots?

Answer

- Variable A: Normal
- Variable B: Right skewed
- Variable C: Right skewed

Tasks for Exercise 04: Missing values

Task 01

```
if (!require("naniar")) install.packages("naniar", dependencies=TRUE)
```

Loading required package: naniar

```
if (!require("mice")) install.packages("mice", dependencies=TRUE)
```

Loading required package: mice

Attaching package: 'mice'

The following object is masked from 'package:stats':

filter

The following objects are masked from 'package:base':

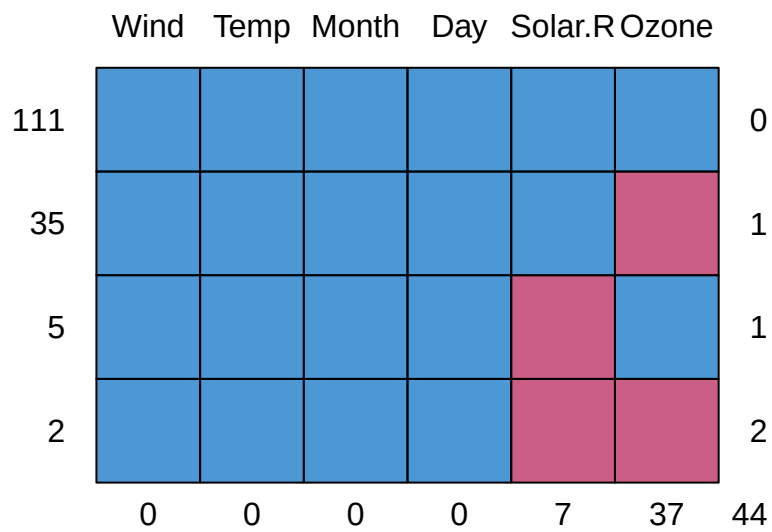
cbind, rbind

```
library(naniar)
library(mice)

head(airquality)
```

	Ozone	Solar.R	Wind	Temp	Month	Day
1	41	190	7.4	67	5	1
2	36	118	8.0	72	5	2
3	12	149	12.6	74	5	3
4	18	313	11.5	62	5	4
5	NA	NA	14.3	56	5	5
6	28	NA	14.9	66	5	6

```
# Missing pattern
md.pattern(airquality)
```



	Wind	Temp	Month	Day	Solar.R	Ozone
111	1	1	1	1	1	0
35	1	1	1	1	1	1
5	1	1	1	1	0	1
2	1	1	1	1	0	2
	0	0	0	0	7	44

```
# Little's test
mcar_test(airquality)
```

```
# A tibble: 1 x 4
  statistic    df p.value missing.patterns
    <dbl> <dbl>   <dbl>         <int>
1     35.1    14 0.00142             4
```

Task 02

```
if (!require("readxl")) install.packages("readxl", dependencies=TRUE)
if (!require("tidyverse")) install.packages("tidyverse", dependencies=TRUE)
```

Loading required package: tidyverse

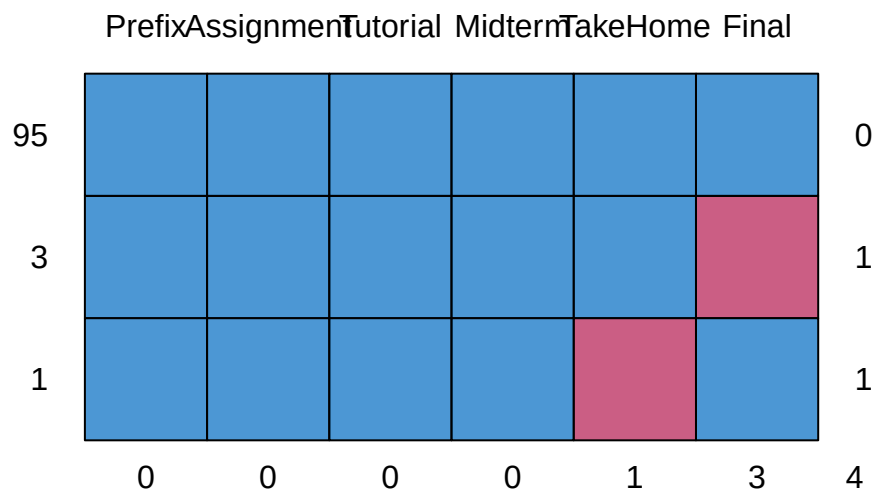
```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks mice::filter(), stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
if (!require("mice")) install.packages("mice", dependencies=TRUE)

library(readxl)
library(mice)
library(tidyverse)

data <- read_excel("/home/nils/dev/mscids-notes/fs26/dq/data/grades.xlsx")

# Missing pattern
md.pattern(data)
```

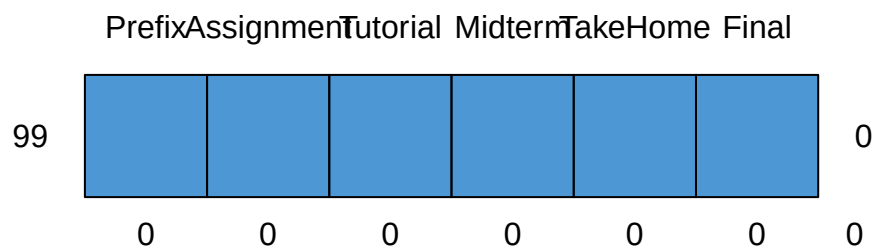


	Prefix	Assignment	Tutorial	Midterm	TakeHome	Final
95	1	1	1	1	1	1 0
3	1	1	1	1	1	0 1
1	1	1	1	1	0	1 1
	0	0	0	0	1	3 4

```
data <- data %>%
  mutate(
    Final = replace_na(Final, mean(Final, na.rm = TRUE)),
    TakeHome = replace_na(TakeHome, mean(TakeHome, na.rm = TRUE))
  )

# Missing pattern after replace
md.pattern(data)
```

```
/\      /\
{  `----'  }
{  0    0  }
==> V <== No need for mice. This data set is completely observed.
\  \|/  /
 `-----'
```



	Prefix	Assignment	Tutorial	Midterm	TakeHome	Final
99	1	1	1	1	1	1 0
	0	0	0	0	0	0 0

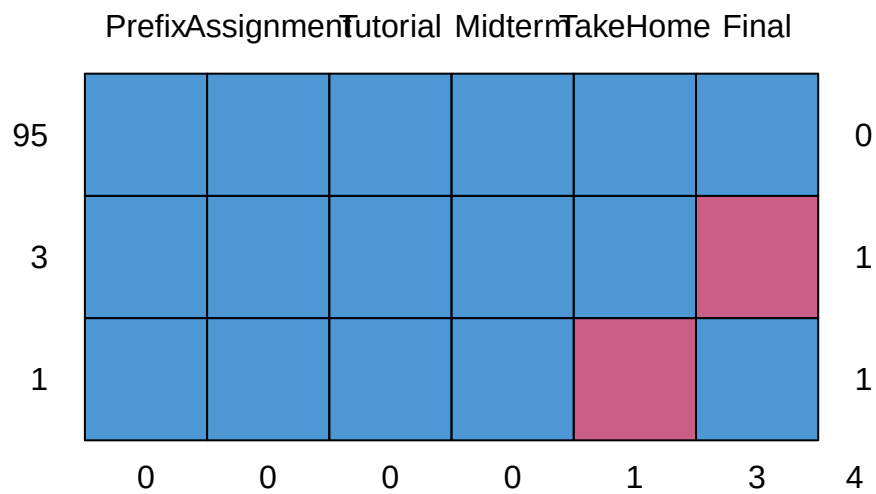
OR

```
if (!require("readxl")) install.packages("readxl", dependencies=TRUE)
if (!require("mice")) install.packages("mice", dependencies=TRUE)

library(readxl)
library(mice)

data <- read_excel("/home/nils/dev/mscids-notes/fs26/dq/data/grades.xlsx")

# Missing pattern
md.pattern(data)
```

	Prefix	Assignment	Tutorial	Midterm	TakeHome	Final	
95	1	1	1	1	1	1	0
3	1	1	1	1	1	0	1
1	1	1	1	1	0	1	1
	0	0	0	0	1	3	4

```
imp_model <- mice(data, method="mean", m=1, maxit=1)
```

```
iter imp variable
1 1 TakeHome Final
```

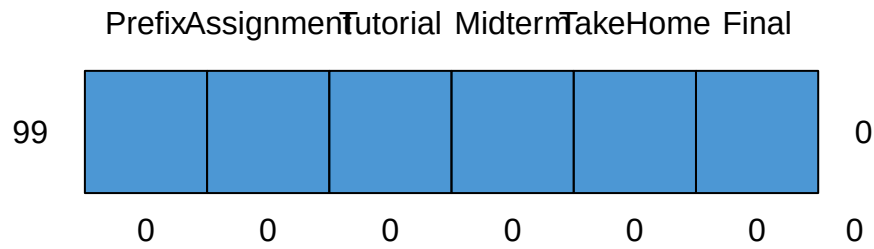
```
data_complete <- complete(imp_model)
```

```
# Missing pattern
md.pattern(data_complete)
```

```
/\      /\
{ `---' }
{ 0 0 }
```

```
==> V <== No need for mice. This data set is completely observed.
```

\ \ / /
 \-----/



	Prefix	Assignment	Tutorial	Midterm	TakeHome	Final
99	1	1	1	1	1	1
	0	0	0	0	0	0

Task 03

Mechanism	Summary
MCAR	A random “glitch” in the system.
MAR	Predictable based on other info you have.
MNAR	The “hidden” value is the reason it’s missing.

Tasks for Exercise 05: Univariate Outlier Detection

Task 01

```

if (!require("readxl")) install.packages("readxl", dependencies=TRUE)
if (!require("dplyr")) install.packages("dplyr", dependencies=TRUE)

library(readxl)
library(dplyr)

data <- read_excel("/home/nils/dev/mscids-notes/fs26/dq/data/HSE.xlsx")
data

```

```

# A tibble: 13,908 x 2
  wtval    sex
  <dbl> <dbl>
1  85.2     1
2  72.7     1
3  84.8     1
4  73.6     1
5  61.3     1
6  81.9     1
7   60     1
8  NA      1
9  NA      1
10 123.     1
# i 13,898 more rows

```

```

data.M <- data %>% filter(sex == 1)
data.F <- data %>% filter(sex == 2)

```

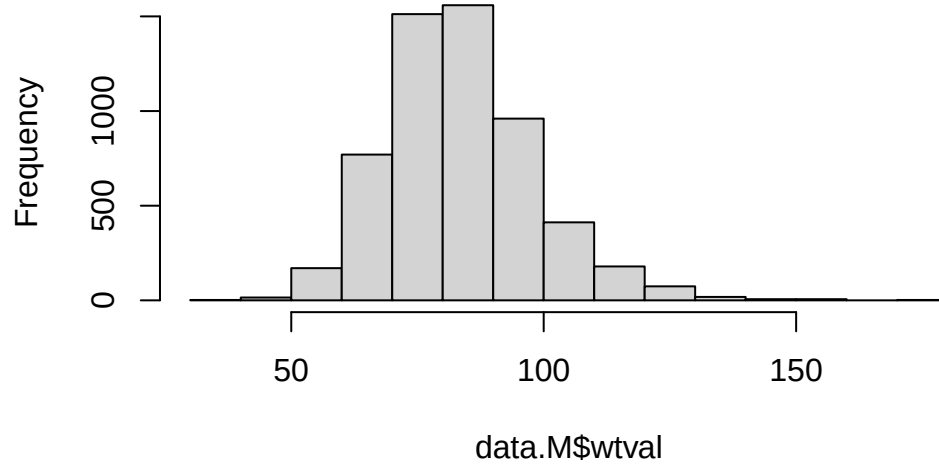
Histogram

```

hist(data.M$wtval)

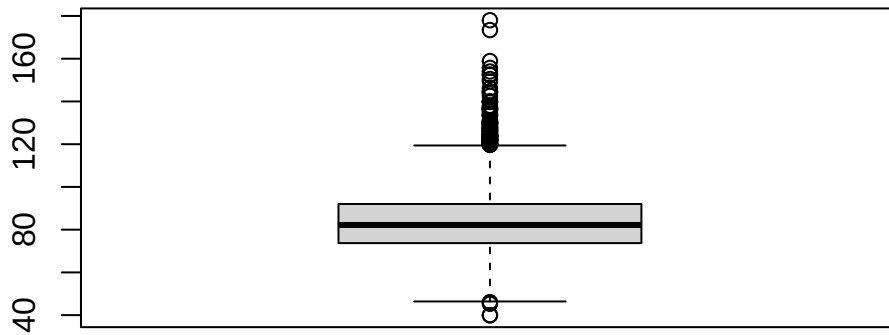
```

Histogram of data.M\$wtval



Boxplot

```
boxplot(data.M$wtval)
```



Percentile Method

```
# Calculate thresholds (removing NA values is crucial here)
lower_bound <- quantile(data.M$wtval, 0.01, na.rm = TRUE)
upper_bound <- quantile(data.M$wtval, 0.99, na.rm = TRUE)

print(lower_bound)
```

```
1%
55.384
```

```
print(upper_bound)
```

```
99%
125.716
```

Z-Score

```

if (!require("dplyr")) install.packages("dplyr", dependencies=TRUE)

library(dplyr)

z_score_data <- data.M %>%
  mutate(
    z_score = (
      wtval - mean(wtval, na.rm = TRUE)) / sd(wtval, na.rm = TRUE)) %>%
  filter(abs(z_score) <= 5)

z_score_data

```

```

# A tibble: 5,682 x 3
   wtval    sex z_score
   <dbl> <dbl>   <dbl>
1  85.2     1  0.105
2  72.7     1 -0.742
3  84.8     1  0.0779
4  73.6     1 -0.681
5  61.3     1 -1.52
6  81.9     1 -0.119
7   60     1 -1.60
8 123.     1  2.69
9  98.4     1  1.000
10 73.3     1 -0.702
# i 5,672 more rows

```

Task 02

Grubbs Test

```

if (!require("readxl")) install.packages("readxl", dependencies=TRUE)
if (!require("outliers")) install.packages("outliers", dependencies=TRUE)

```

Loading required package: outliers

```
library(readxl)
library(outliers)

data <- read_excel("/home/nils/dev/mscids-notes/fs26/dq/data/HSE.xlsx")

# Grubbs' test for the maximum value
grubbs_result <- grubbs.test(data$wtval)
print(grubbs_result)
```

Grubbs test for one outlier

```
data: data$wtval
G = 6.61164, U = 0.99651, p-value = 2.29e-07
alternative hypothesis: highest value 184.3 is an outlier
```

Rosner's Test

```
if (!require("EnvStats")) install.packages("EnvStats", dependencies=TRUE)
```

Loading required package: EnvStats

Attaching package: 'EnvStats'

The following objects are masked from 'package:stats':

```
predict, predict.lm
```

The following object is masked from 'package:base':

```
print.default
```

```
library(EnvStats)

# Rosner's test checking for up to 10 potential outliers
rosner_result <- rosnerTest(data$wtval, k = 10)
```

Warning in `rosnerTest(data$wtval, k = 10)`: 1389 observations with NA/NaN/Inf in 'x' removed.

```
print(rosner_result$all.stats)
```

	i	Mean.i	SD.i	Value	Obs.Num	R.i+1	lambda.i+1	Outlier
1	0	76.28451	16.33717	184.3	12420	6.611640	4.609832	TRUE
2	1	76.27588	16.30927	178.0	5562	6.237196	4.609815	TRUE
3	2	76.26776	16.28455	173.4	4618	5.964685	4.609798	TRUE
4	3	76.26000	16.26204	158.9	3876	5.081773	4.609782	TRUE
5	4	76.25339	16.24590	158.9	6366	5.087228	4.609765	TRUE
6	5	76.24679	16.22974	156.2	13068	4.926339	4.609748	TRUE
7	6	76.24040	16.21464	155.7	5196	4.900485	4.609731	TRUE
8	7	76.23405	16.19972	155.7	8309	4.905391	4.609715	TRUE
9	8	76.22770	16.18478	154.0	1929	4.805274	4.609698	TRUE
10	9	76.22148	16.17048	152.5	3223	4.717146	4.609681	TRUE

Tasks for Exercise 06: Multivariate Outlier detection – Cluster Analysis

Change the parameter `centers` to compare different group sizes.

```
if (!require("readxl")) install.packages("readxl", dependencies=TRUE)
if (!require("MVA")) install.packages("MVA", dependencies=TRUE)
```

Loading required package: MVA

Loading required package: HSAUR2

Loading required package: tools

Attaching package: 'HSAUR2'

The following object is masked from 'package:tidyr':

household

The following object is masked from 'package:mice':

toenail

```
if (!require("cluster")) install.packages("cluster", dependencies=TRUE)
```

Loading required package: cluster

```
library(readxl)
library(MVA)
library(cluster)

data <- read_excel("/home/nils/dev/mscids-notes/fs26/dq/data/data_extreme.xlsx")
head(data)
```

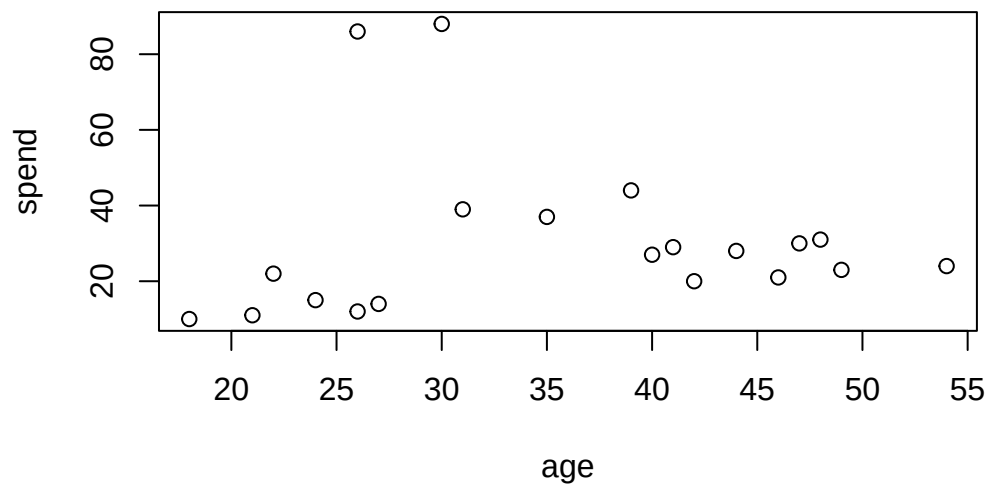
```
# A tibble: 6 x 2
  age spend
<dbl> <dbl>
1    18    10
2    21    11
3    22    22
4    24    15
5    26    12
6    26    86
```

data

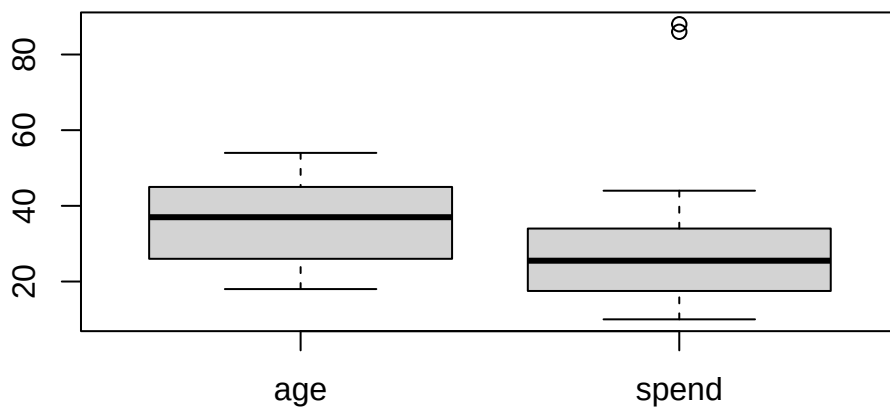
```
# A tibble: 20 x 2
  age spend
<dbl> <dbl>
1    18    10
2    21    11
3    22    22
4    24    15
5    26    12
6    26    86
7    27    14
8    30    88
9    31    39
10   35    37
```

11	39	44
12	40	27
13	41	29
14	42	20
15	44	28
16	46	21
17	47	30
18	48	31
19	49	23
20	54	24

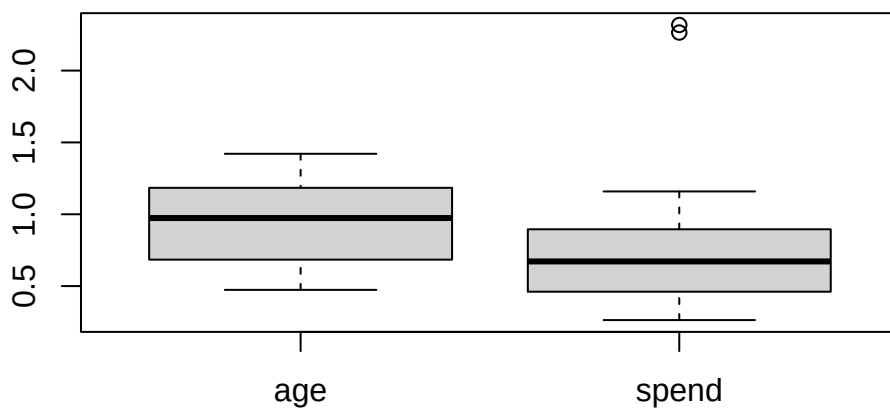
```
plot(data)
```



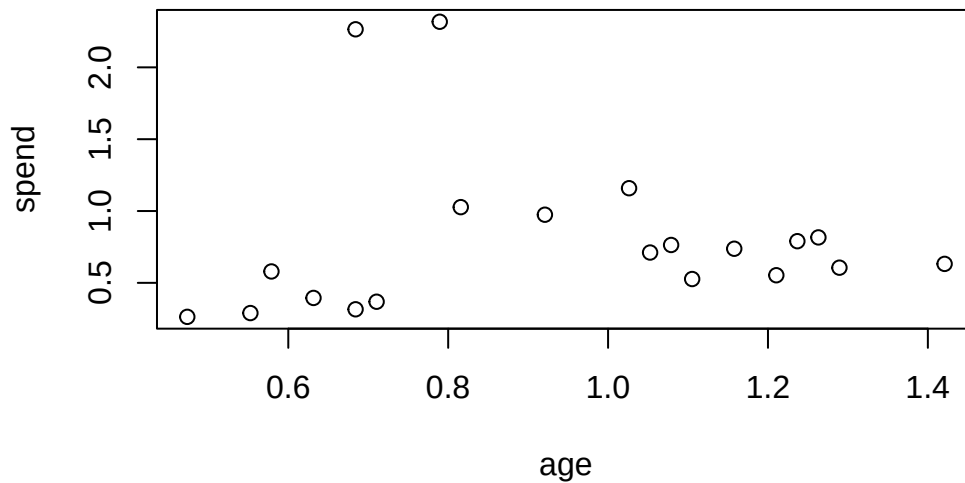
```
boxplot(data)
```



```
datas <- scale(data, center = FALSE, scale = TRUE)
boxplot(datas)
```



```
plot(datas)
```



```
dists <- dist(datas)
```

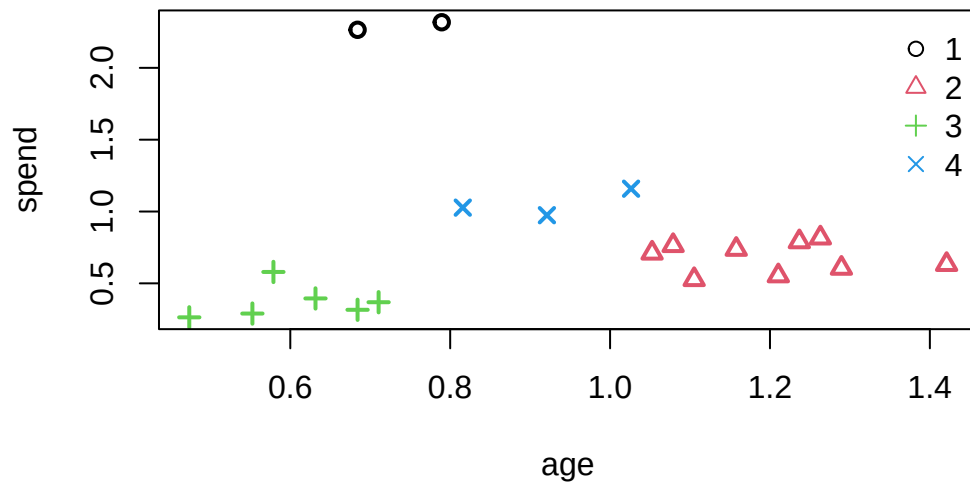
```
km <- kmeans(datas, centers = 4, nstart = 10)
groups_km <- km$cluster
groups_km
```

```
[1] 3 3 3 3 3 1 3 1 4 4 4 2 2 2 2 2 2 2 2 2
```

```
cluster_size <- cbind(sum(groups_km == 1), sum(groups_km == 2), sum(groups_km == 3), sum(groups_km == 4))
cluster_size
```

```
      [,1] [,2] [,3] [,4]
[1,]     2     9     6     3
```

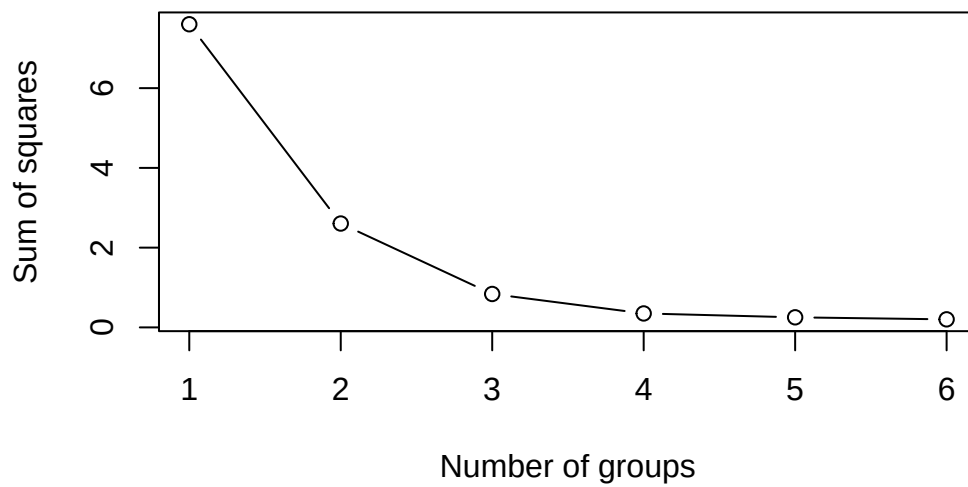
```
plot(datas, pch = groups_km, col=groups_km, lwd=2)
legend("topright", legend = 1:4, pch = 1:4, col=1:4, bty="n")
```



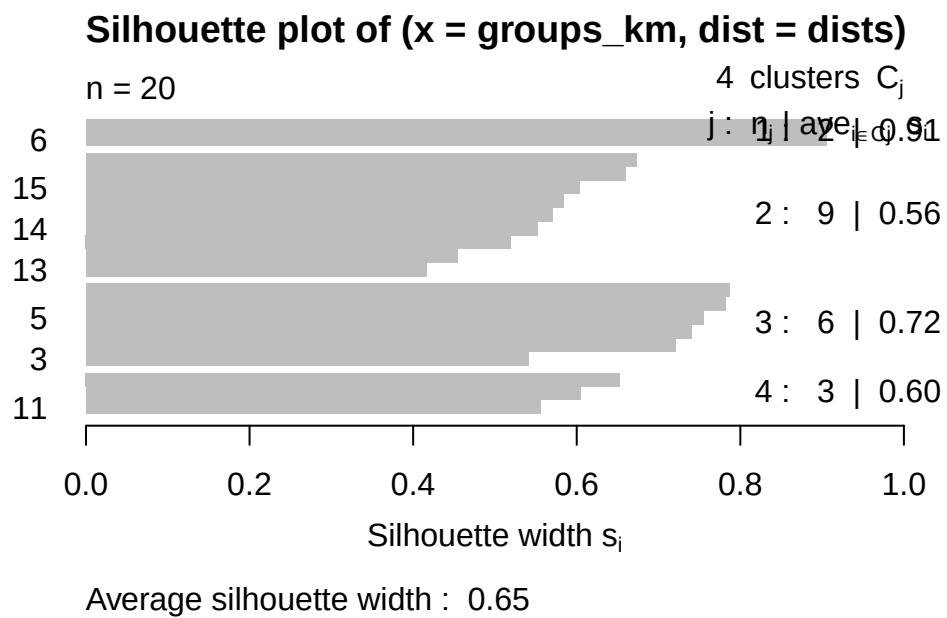
```

reps <- rep(0, 6)
for (i in 1:6) reps[i] <- sum(kmeans(datas, centers = i, nstart = 20)$withinss)
par(mfrow = c(1,1))
plot(1:6, reps, type = "b", xlab = "Number of groups", ylab = "Sum of squares")

```



```
plot(silhouette(groups_km, dists))
```



Tasks for Exercise 07: Multivariate Outlier Detection – Other Algorithms

Task 01

```
library(readxl)
library(questionr)

data <- read_excel("/home/nils/dev/mscids-notes/fs26/dq/data/learning.xlsx")

data <- rename.variable(data, "effort", "x")
data <- rename.variable(data, "success", "y")
plot(data$x, data$y, main="RANSAC Fitting") # explicitly plotting x and y

ransac <- function(data, n, k, t, d) {
  iterations <- 0
  bestfit <- NULL
  besterr <- 1e5

  while (iterations < k) {
    maybeinliers <- sample(nrow(data), n)
    maybemodel <- lm(y ~ x, data = data, subset = maybeinliers)

    # Skip if model failed to find valid coefficients (e.g., vertical line)
    if (any(is.na(coef(maybemodel)))) {
      iterations <- iterations + 1
      next
    }

    alsoinliers <- NULL
    m <- maybemodel$coefficients[2] # slope
    c <- maybemodel$coefficients[1] # intercept

    for (point in setdiff(1:nrow(data), maybeinliers)) {
      # Use [[ ]] to ensure we get a single numeric value, not a 1x1 tibble
      x0 <- data[[point, 1]]
      y0 <- data[[point, 2]]

      # Corrected distance formula ( $m^2 + 1$ )
      dist <- abs(m * x0 - y0 + c) / sqrt(m^2 + 1)
```

```

    # Safety check against NA distances
    if (!is.na(dist) && dist < t) {
      alsoinliers <- c(alsoinliers, point)
    }
  }

  if (length(alsoinliers) > d) {
    bettermodel <- lm(y ~ x, data = data, subset = c(maybeinliers, alsoinliers))
    thiserr <- summary(bettermodel)$sigma
    if (thiserr < besterr) {
      bestfit <- bettermodel
      besterr <- thiserr
    }
  }
  iterations <- iterations + 1
}
return(bestfit)
}

# Base OLS model
abline(lm(y ~ x, data = data), col = "red", lwd = 2)

x <- 1
repeat {
  x <- x + 1

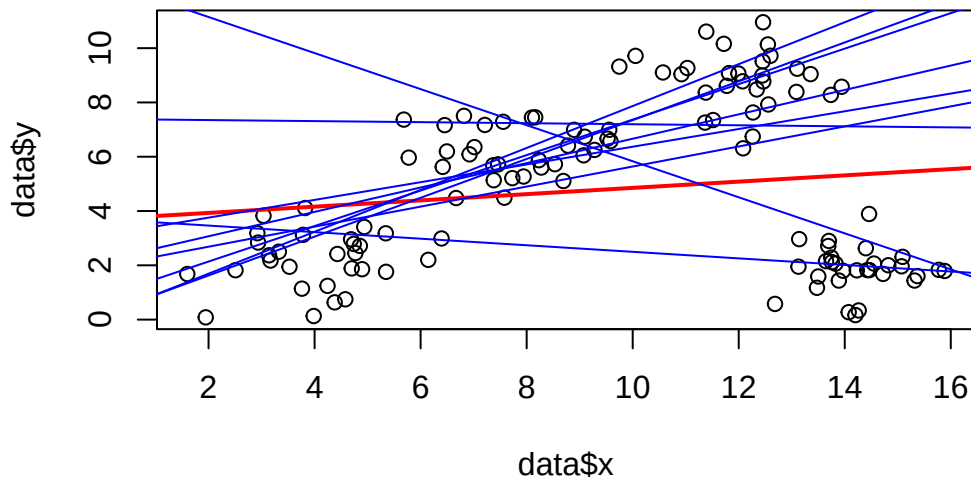
  # Run RANSAC and capture the result
  r_model <- ransac(data, n = 10, k = 10, t = 0.5, d = 10)

  # Only attempt to plot if RANSAC successfully found a model
  if (!is.null(r_model)) {
    abline(r_model, col = "blue")
  } else {
    message("Iteration ", x, ": RANSAC could not find a model meeting threshold 'd'.")
  }

  if (x == 10) { break }
  # Note: I removed q() here. Having q() directly after a loop in Quarto
  # will terminate the R session mid-render, which will also cause a crash.
}

```


RANSAC Fitting



Task 02

```
library(readxl)
library(dbscan)
```

Attaching package: 'dbscan'

The following object is masked from 'package:stats':

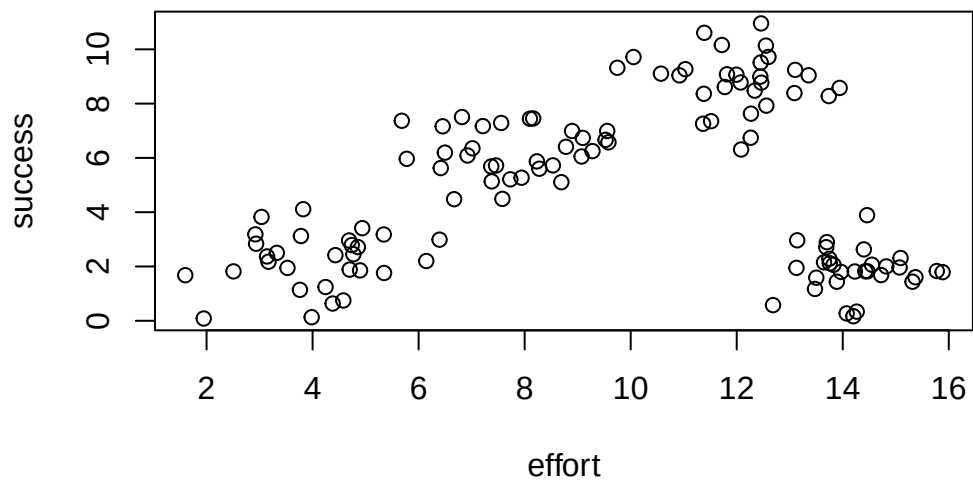
as.dendrogram

```
data <- read_excel("/home/nils/dev/mscids-notes/fs26/dq/data/learning.xlsx")
head(data)
```

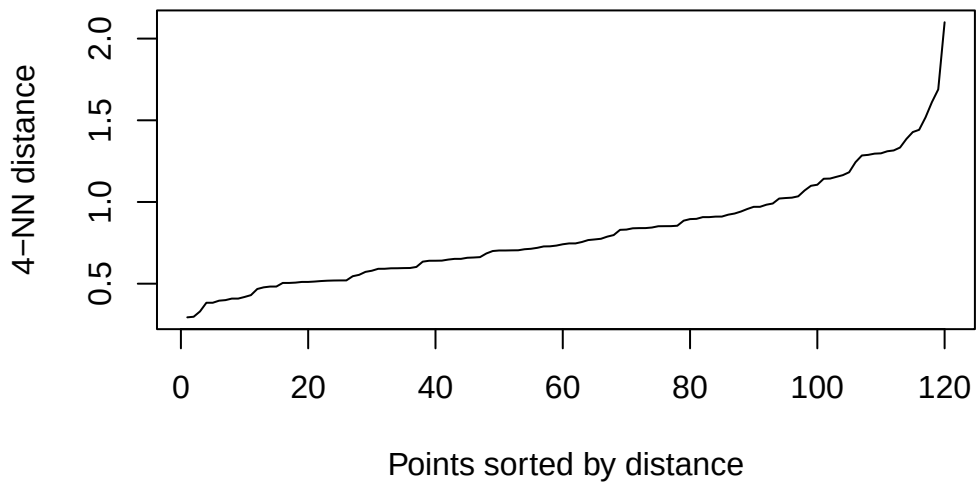
```
# A tibble: 6 x 2
  effort success
  <dbl>   <dbl>
1     1.0     0.0
2     1.0     0.0
3     1.0     0.0
4     1.0     0.0
5     1.0     0.0
6     1.0     0.0
```

1	2.93	2.84
2	3.82	4.11
3	4.69	2.96
4	4.38	0.635
5	4.58	0.748
6	6.39	2.99

```
plot(data)
```



```
kNNdistplot(data, k = 4)
```



```
dbscanResult <- dbscan(data, eps= 15, minPts=4)
dbscanResult
```

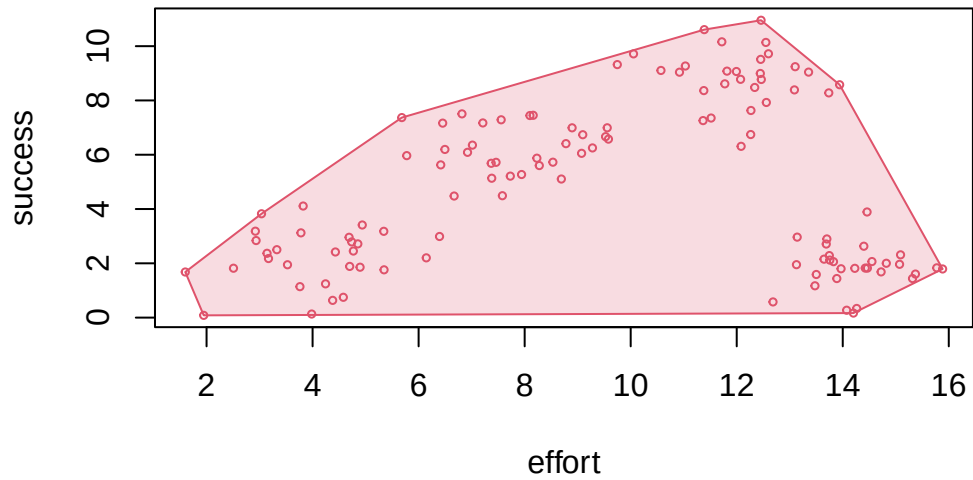
DBSCAN clustering for 120 objects.
Parameters: eps = 15, minPts = 4
Using euclidean distances and borderpoints = TRUE
The clustering contains 1 cluster(s) and 0 noise points.

```
1
120
```

Available fields: cluster, eps, minPts, metric, borderPoints

```
hullplot(data, dbscanResult)
```

Convex Cluster Hulls



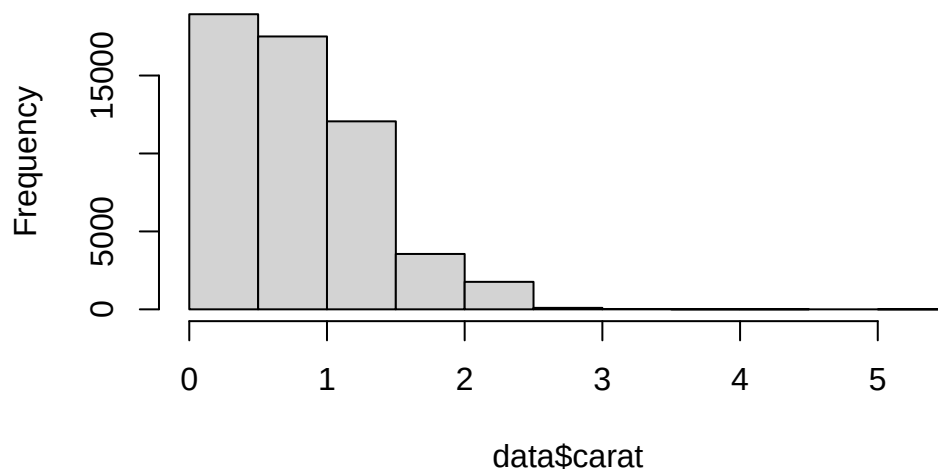
Tasks for Exercise 10

Task 01

```
library(ggplot2)
data <- diamonds

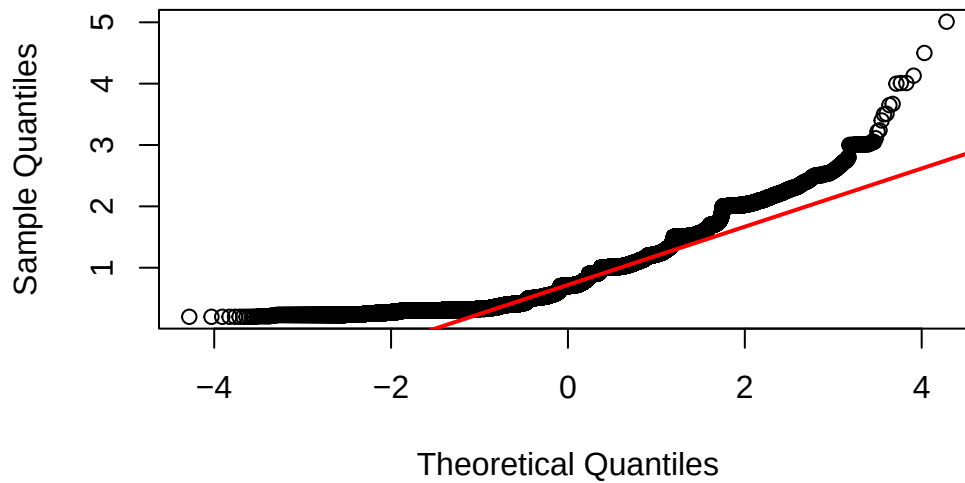
hist(data$carat)
```

Histogram of data\$carat

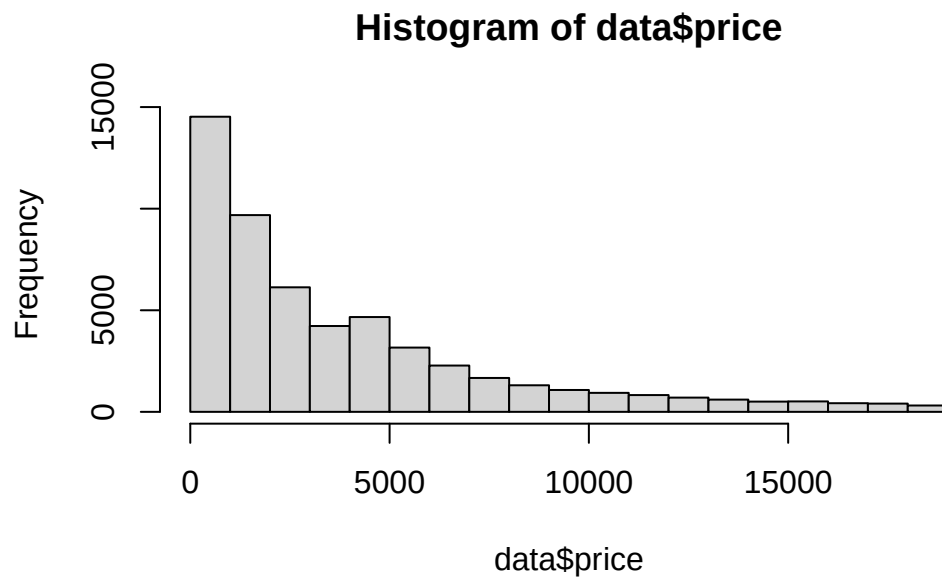


```
qqnorm(data$carat)
qqline(data$carat, col = "red", lwd = 2)
```

Normal Q-Q Plot

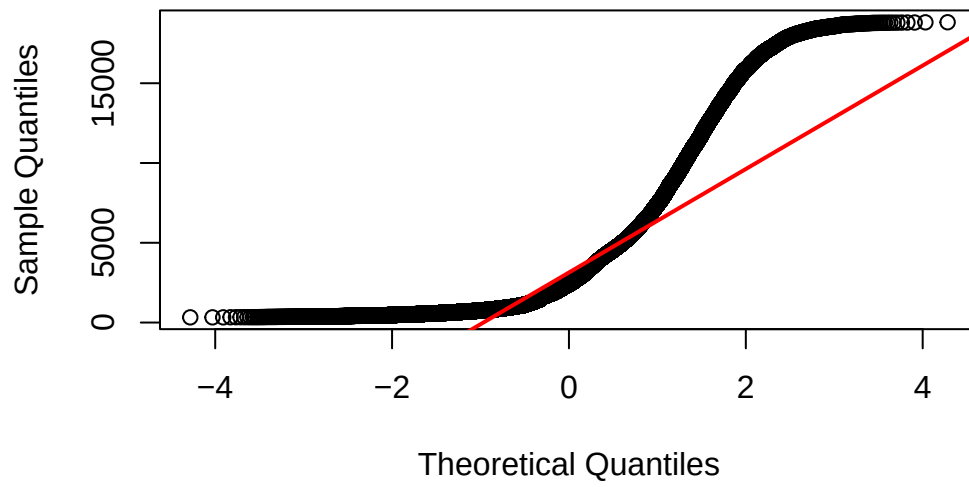


```
hist(data$price)
```



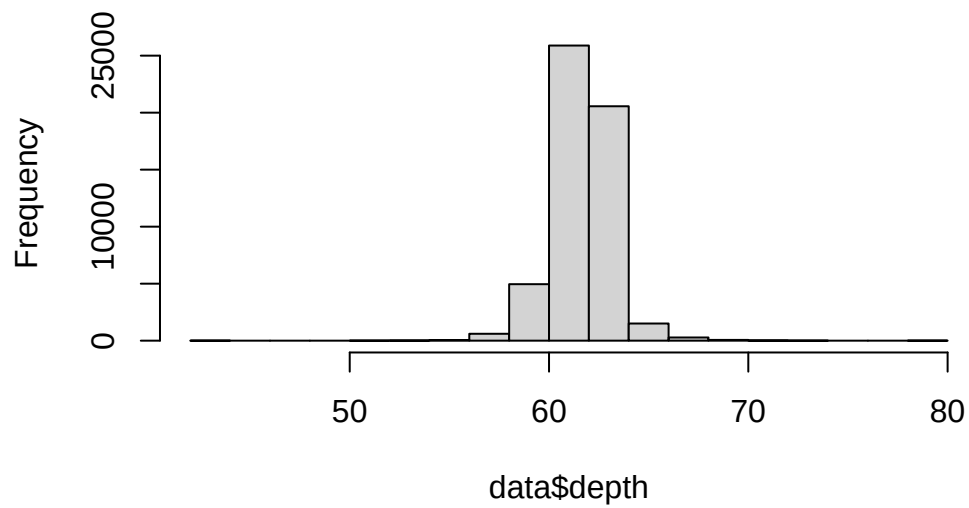
```
qqnorm(data$price)  
qqline(data$price, col = "red", lwd = 2)
```

Normal Q-Q Plot



```
hist(data$depth)
```

Histogram of data\$depth



```
qqnorm(data$depth)
qqline(data$depth, col = "red", lwd = 2)
```

